

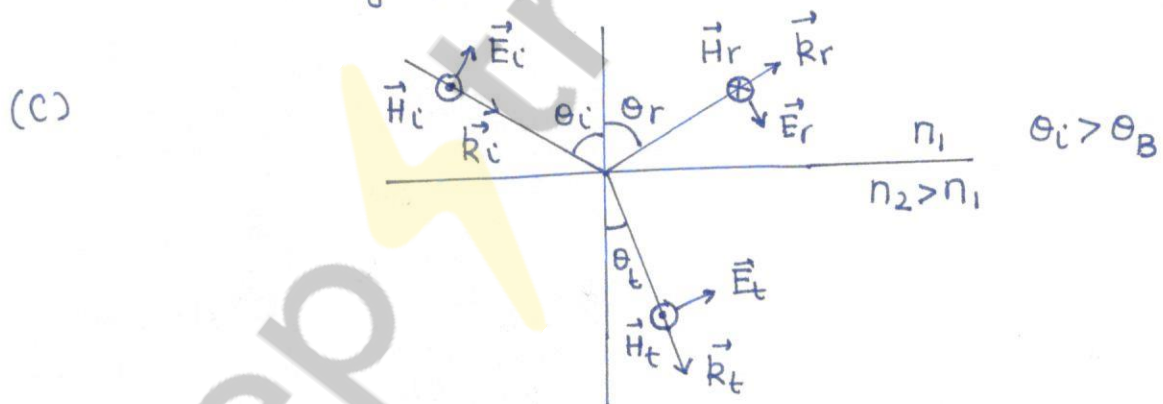
Solution
PH102 (End-Sem)
(2023)

$$1 (a) \quad \vec{\nabla} \times \vec{E} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ r\cos^2\theta & -r^2\cos\theta\sin\theta & 3r^2\sin\theta \end{vmatrix}$$

$$= 3 \cot \theta \hat{r} - 6 \hat{\theta} \neq 0$$

Hence, $\vec{E}$ is not a possible E.F. ①

(b) (iii) Reflected and transmitted waves are mutually perpendicular. ①

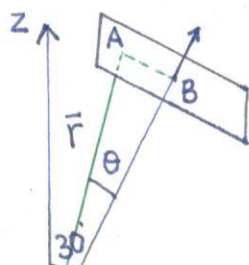


$$(d) \quad \vec{E} = \vec{E}_0 \exp i \left[\omega t - \frac{kx}{2} - \frac{\sqrt{3}kz}{2} \right]$$

$$\vec{k} \cdot \vec{r} = \frac{kx}{2} + \frac{\sqrt{3}kz}{2} = \left(\frac{k}{2} \hat{x} + \frac{\sqrt{3}k}{2} \hat{z} \right) \cdot (x\hat{x} + z\hat{z})$$

$$\Rightarrow \vec{k} = \frac{k}{2} \hat{x} + \frac{\sqrt{3}k}{2} \hat{z}$$

$\Rightarrow$ wave is travelling in a direction 60° from x-axis and 30° from z-axis and 90° from y-axis ①

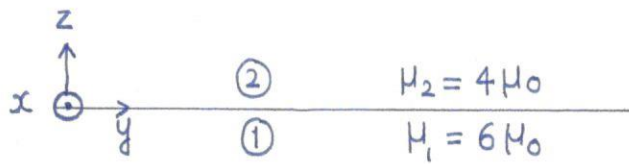


$$\vec{k} \cdot \vec{r} = |\vec{k}| |\vec{r}| \cos \theta$$

$$= k \times OB = \text{const}$$

which is an equation of plane $\rightarrow$ plane wave.

(e)



$$\vec{B}_2 = 5\hat{x} + 8\hat{z} \quad \frac{mWb}{m^2}$$

$$\vec{K}_f = \frac{1}{\mu_0} \hat{y} \quad \frac{mA}{m}$$

$$\vec{H}_2 = \frac{\vec{B}_2}{\mu_2} = \frac{1}{4\mu_0} (5\hat{x} + 8\hat{z}) \quad \frac{mA}{m}$$

(1/2)

$$\text{Let } \vec{B}_1 = B_{x1}\hat{x} + B_{y1}\hat{y} + B_{z1}\hat{z}$$

$$\therefore B_1^\perp = B_2^\perp \Rightarrow B_{z1} = 8$$

(1/2)

$$\text{Also } \frac{\vec{B}_1^\parallel}{\mu_1} - \frac{\vec{B}_2^\parallel}{\mu_2} = \vec{K}_f \times \hat{z}$$

(1/2)

$$\therefore \frac{1}{6\mu_0} (B_{1x}\hat{x} + B_{1y}\hat{y}) - \frac{1}{4\mu_0} (5\hat{x}) = \frac{1}{\mu_0} \hat{x}$$

$$\rightarrow B_{1y} = 0 ; B_{1x} = \frac{27}{2}$$

$$\therefore \vec{B}_1 = \left(\frac{27}{2} \hat{x} + 8\hat{z} \right) \frac{mWb}{m^2}$$

(1)

$$\vec{H}_1 = \frac{\vec{B}_1}{\mu_1} = \frac{1}{\mu_0} \left(\frac{9}{4} \hat{x} + \frac{4}{3} \hat{z} \right) \frac{mA}{m}$$

(1/2)



Problem 2. Two long coaxial cylindrical metal tubes (inner radius a , outer radius b) stand vertically in a tank of dielectric oil (susceptibility χ_e , mass density ρ). The inner one is maintained at potential V , and the outer one is grounded (Fig. 2). To what height (h) does the oil rise, in the space between the tubes?

Solution: Air part:

$$\vec{E}_a = \frac{Q}{2\pi\epsilon_0 r L} \hat{r} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

$$Q = \lambda L$$

$$\vec{E}_a = \frac{V}{\ln(b/a)} \frac{\hat{r}}{r} \quad [1]$$

$$V = V_b - V_a = \int \vec{E}_a \cdot d\vec{l}$$

$$= \int_a^b \frac{\lambda}{2\pi\epsilon_0 r} dr$$

$$= \frac{\lambda}{2\pi\epsilon_0} \ln(b/a)$$

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln(b/a) = \frac{\chi}{2\pi\epsilon} \ln(b/a)$$

$$\frac{\lambda}{\epsilon_0} = \frac{\chi}{\epsilon} \Rightarrow \chi' = \frac{\epsilon}{\epsilon_0} \lambda = \epsilon_r \lambda$$

$$W = \frac{1}{2} \int_a^b \int_0^{2\pi} \int_0^h \epsilon E_a^2 dz + \frac{1}{2} \int_a^b \int_0^{2\pi} \int_h^L \epsilon_0 E_a^2 dz$$

$$= \frac{\epsilon}{2} \frac{V^2}{\left\{ \ln(b/a) \right\}^2} \int_a^b \int_0^{2\pi} \int_0^h \frac{1}{r^2} r dr d\phi dz + \frac{\epsilon_0}{2} \frac{V^2}{\left\{ \ln(b/a) \right\}^2} \int_a^b \int_0^{2\pi} \int_h^L \frac{1}{r^2} r dr d\phi dz$$

$$= \frac{\epsilon}{2} \frac{V^2}{\left\{ \ln(b/a) \right\}^2} \ln(b/a) 2\pi h + \frac{\epsilon_0}{2} \frac{V^2}{\left\{ \ln(b/a) \right\}^2} \ln(b/a) 2\pi (L-h)$$

$$= \frac{\pi V^2}{\ln(b/a)} \left\{ \epsilon h + \epsilon_0 (L-h) \right\}$$

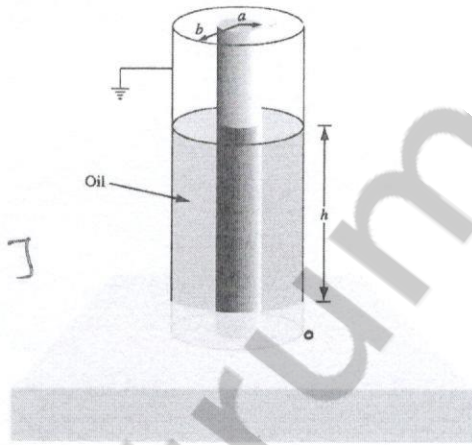
$$\sum \ln x^2 \neq 2 \ln x = \ln x^2$$

The vertical electrical force on the oil may be equated the gravitational force

$$F_z = \frac{\partial W}{\partial h} = \frac{\pi V^2}{\ln(b/a)} \left\{ \epsilon - \epsilon_0 \right\} = \pi (b^2 - a^2) z \rho g \quad [1]$$

$$z = \frac{\pi V^2 (\epsilon - \epsilon_0)}{\pi (b^2 - a^2) \ln(b/a) \rho g} = \frac{V^2 (\epsilon - \epsilon_0)}{\rho (b^2 - a^2) g \ln(b/a)} = \frac{\epsilon_0 (\epsilon/\epsilon_0 - 1) V^2}{\rho (b^2 - a^2) g \ln(b/a)}$$

$$= \frac{\epsilon_0 \chi_e V^2}{\rho (b^2 - a^2) g \ln(b/a)} \quad [1]$$



Oil part: $\vec{E}_o$

$$\oint \vec{D} \cdot d\vec{l} = Q_{enc}$$

$$\epsilon \vec{E}_o = \frac{Q_{enc}}{2\pi r L} \hat{r}$$

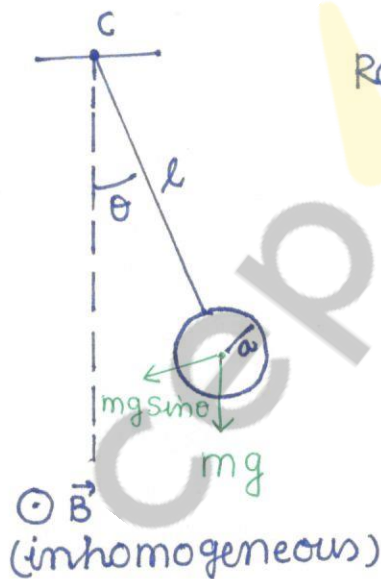
$$\vec{E}_o = \frac{\chi'}{2\pi\epsilon r} \hat{r} = \frac{V}{\ln(b/a)} \frac{\hat{r}}{r} \quad [1]$$

$$V = V_b - V_a = \int \vec{E}_o \cdot d\vec{l}$$

$$= \int_a^b \frac{\chi'}{2\pi\epsilon r} dr$$

$$= \frac{\chi'}{2\pi\epsilon} \ln(b/a)$$

3(a)



Rate of Energy dissipated:

$$\tau \omega = \frac{\epsilon^2}{R} \quad \tau: \text{torque} \quad (1)$$

$$\epsilon = -\frac{d\Phi}{dt} = -\pi a^2 \frac{dB_{\perp}}{dt}$$

$$= -\pi a^2 \frac{dB_{\perp}}{d\theta} \frac{d\theta}{dt}$$

$$= -\pi a^2 \omega \frac{dB_{\perp}}{d\theta} \quad (1)$$

$$\text{Dissipative torque } \tau = (\pi a^2)^2 \left(\frac{dB_{\perp}}{d\theta} \right)^2 \frac{\omega}{R} \quad (1/2)$$

$$\text{Moment of inertia about point 'c' of the loop} \\ = m[(l+a)^2 + a^2]^* \quad (1/2)$$

 $\therefore$ Equation of motion is

$$m[a^2 + (l+a)^2] \frac{d^2\theta}{dt^2} + \frac{(\pi a^2)^2}{R} \left(\frac{dB_{\perp}}{d\theta} \right)^2 \frac{d\theta}{dt} + mg(l+a) \sin \theta = 0 \quad (1)$$



3) b)

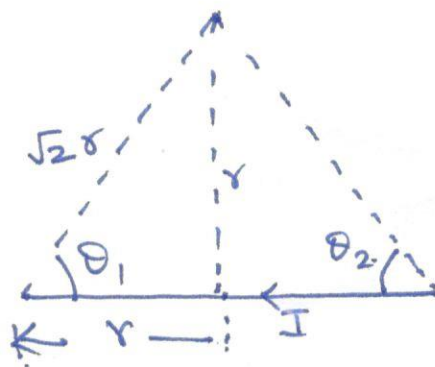
Field due to the straight wire

$$B_1 = \frac{\mu_0 I}{4\pi r} (\cos\theta_2 + \cos\theta_1)$$

$$\cos\theta_2 = \cos\theta_1 = \frac{r}{\sqrt{2}r} = \frac{1}{\sqrt{2}}$$

$$\therefore \vec{B}_1 = \frac{\mu_0 I}{2\sqrt{2}\pi r} \hat{z}$$

(1)



Field due to the arc:

$$\vec{r}' = r \cos\phi \hat{x} - r \sin\phi \hat{y}$$

$$\vec{r} = r \hat{y}$$

$$\vec{r}'' = \vec{r} - \vec{r}'$$

$$= -r \cos\phi \hat{x} + r(1 + \sin\phi) \hat{y}$$

$$d\vec{r} = r \sin\phi d\phi \hat{x} + r \cos\phi d\phi \hat{y}$$

$$d\vec{r} \times \vec{r}'' = r d\phi [1 + \sin\phi] \hat{z}$$

$$\therefore \vec{B}_2 = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{r} \times \vec{r}''}{(\vec{r}'')^3}$$

$$= \frac{\mu_0 I}{4\pi} r^2 \hat{z} \int_0^\pi \frac{(1 + \sin\phi) d\phi}{[r^2 \cos^2\phi + r^2 (1 + \sin^2\phi)]^{3/2}}$$

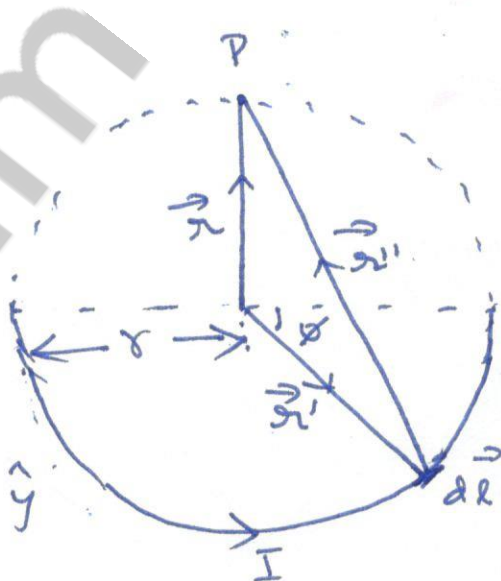
$$= \frac{\mu_0 I r^2}{4\pi (2r^2)^{3/2}} \hat{z} \int_0^\pi \frac{(1 + \sin\phi) d\phi}{(1 + \sin\phi)^{3/2}}$$

$$= \frac{\mu_0 I}{8\sqrt{2}\pi r} \hat{z} \int_0^\pi \frac{d\phi}{\sqrt{1 + \sin\phi}}$$

$$= \frac{\mu_0 I}{8\sqrt{2}\pi r} \hat{z} 2\sqrt{2} \tanh^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= \frac{\mu_0 I}{4\pi r} \tanh^{-1}\left(\frac{1}{\sqrt{2}}\right) \hat{z}$$

(2)

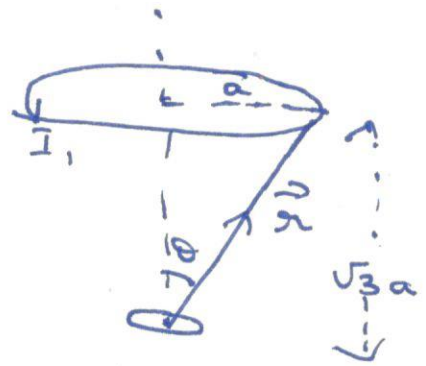


Field at P is the vector sum of fields due to the straight wire and the arc (4)

4(a) (i) Vector potential due to the magnetic dipole

$$\vec{A} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^2}$$

$$= \frac{\mu_0 I}{4\pi} \frac{I_1 ds \sin\theta}{r^2} \hat{\phi}$$



Here $r^2 = a^2 + (\sqrt{3}a)^2$, $\sin\theta = \frac{a}{r} = \frac{a}{[a^2 + (\sqrt{3}a)^2]^{1/2}}$

$$\therefore \vec{A} = \frac{\mu_0}{32\pi} \frac{I_1 ds}{a^2} \hat{\phi} \quad (1)$$

$$\oint \vec{A} \cdot d\vec{l} = \int \vec{B} \cdot d\vec{a} = \Phi \quad (\text{flux})$$

$$\Rightarrow A(2\pi a) = \Phi$$

$$\therefore \Phi = \frac{\mu_0}{4\pi} \frac{I_1 ds a (2\pi a)}{[a^2 + (\sqrt{3}a)^2]^{1/2}}$$

$$= \frac{\mu_0 I_1 ds}{16a} \quad (1)$$

Mutual inductance $M = \frac{\Phi}{I_1} = \frac{\mu_0 ds}{16a} \quad (1/2)$

(ii) Magnetic field at the axis of a current carrying loop $\vec{B} = \frac{\mu_0 I}{2} \frac{a^2}{(a^2 + z^2)^{3/2}} \hat{z} \quad (1)$

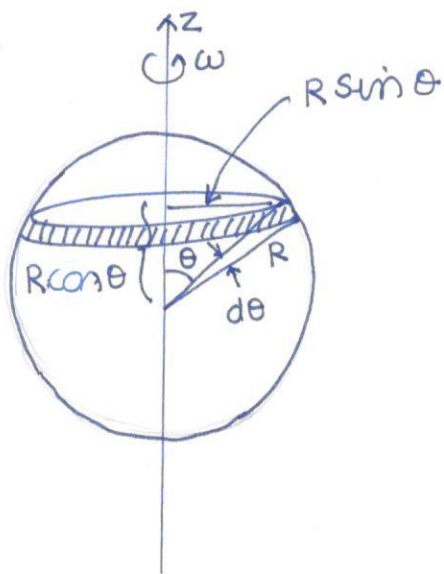
Here $I = I_2$, $z = \sqrt{3}a$

$$\therefore \vec{B} = \frac{\mu_0 I_2}{16a} \hat{z}$$

$$\Phi = \int \vec{B} \cdot d\vec{a} = \frac{\mu_0 I_2 ds}{16a} \quad (1)$$

$$M = \Phi / I_2 = \frac{\mu_0 ds}{16a} \quad (1/2)$$

4(b)



Magnetic field at a distance z above the center of the loop is

$$\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}} \hat{z}$$

For the element

$$d\vec{B} = \frac{\mu_0 I}{2} \frac{R^2 \sin^2 \theta \hat{z}}{[R^2 \sin^2 \theta + R^2 (1 - \cos \theta)^2]^{3/2}}$$

$$\therefore R \rightarrow R \sin \theta \quad z \rightarrow R - R \cos \theta \quad (1/2)$$

Total charge on the ring $dq = \sigma(2\pi R \sin \theta) R d\theta$

Time of one revolution $dt = 2\pi/\omega$

$$\therefore I = \frac{dq}{dt} = \sigma \omega R^2 \sin \theta d\theta$$

$$\therefore d\vec{B} = \frac{\mu_0}{2} \frac{\sigma \omega R^2 \sin\theta d\theta R^2 \sin^2\theta}{R^3 [2(1-\cos\theta)]^{3/2}} \hat{z}$$

$$\therefore \vec{B} = \hat{z} \frac{\mu_0 \sigma \omega R}{4\sqrt{2}} \int_0^{\pi} \frac{1 - \cos^2 \theta}{(1 - \cos \theta)^{3/2}} \sin \theta d\theta$$

$$= \hat{z} \frac{\mu_0 \sigma \omega R}{4\sqrt{2}} \frac{8\sqrt{2}}{3} = \frac{2}{3} \mu_0 \sigma \omega R \hat{z} \quad \triangle (1)$$

5(a)

At time t ,

current in outer solenoid $I = kt$ ($k: \text{const}$)
 in inner solenoid $2I = 2kt$

$$B_{\text{inside outer solenoid}} = \mu_0 n k t = B \quad \text{--- (1)}$$

$$B_{\text{inside inner solenoid}} = 3\mu_0 n k t = B + 2B \quad (1)$$

magnetic flux enclosed by particle's trajectory:

$$\Phi = \pi R^2 (2B) + \pi r^2 (B)$$

(1/2)

Induced (const) electric field:

$$E (2\pi r) = \frac{d\Phi}{dt} = (2R^2 + r^2) \pi \mu_0 n k$$

$$\therefore E = \left(\frac{2R^2 + r^2}{r} \right) \frac{\mu_0 n k}{2} \quad \text{--- (2) (1)}$$

$\vec{B}$ keeps charged particle in it's circular orbit

hence, $\frac{mv^2}{r} = qvB$

(1/2)

But tangential force: $ma = qE$

(1/2)

$$\therefore v = at \quad (\because u = 0)$$

$$= \left(\frac{2R^2 + r^2}{r} \right) \frac{\mu_0 n k}{2} \frac{q}{m} t \quad \text{--- (3) (1/2)}$$

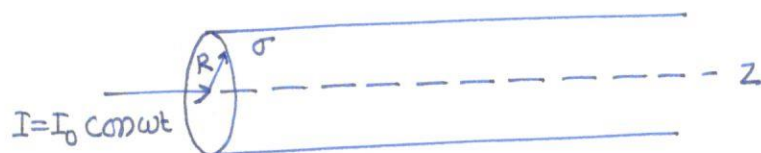
Using (1), (2), (3)

$$\frac{m}{r} \left(\frac{2R^2 + r^2}{r} \right) \frac{\mu_0 n k}{2} \frac{q}{m} t = q \mu_0 n k t$$

$$\Rightarrow r = \sqrt{2} R$$

①
 Δ

5(b)



$$\textcircled{1} \quad \vec{E} = \frac{\vec{J}}{\sigma} = \hat{z} \frac{I_0 \cos \omega t}{\pi R^2 \sigma}$$

(1/2)

$$\textcircled{2} \quad B(2\pi R) = \mu_0 I_0 \cos \omega t$$

$$\rightarrow \vec{B} = \hat{\phi} \frac{\mu_0 I_0 \cos \omega t}{2\pi R}$$

(1/2)

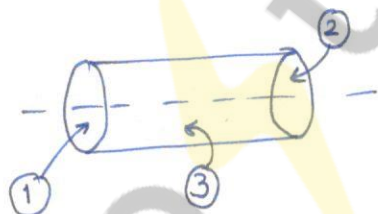
$$\textcircled{3} \quad \vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = -\frac{1}{\mu_0} \frac{\mu_0 I_0^2 \cos^2 \omega t}{2\pi^2 R^3 \sigma} \hat{r}$$

(1/2)

$$\therefore \langle \vec{S} \rangle_t = -\frac{I_0^2 \mu_0}{4\pi^2 R^3 \sigma \mu_0} \hat{r}$$

(1/2)

④ consider unit length of the cylinder



Energy passing through the surface:

$$\oint \langle \vec{S} \rangle_t \cdot d\vec{a} = \int_{(1)} \langle \vec{S} \rangle_t \cdot d\vec{a} + \int_{(2)} \langle \vec{S} \rangle_t \cdot d\vec{a} + \int_{(3)} \langle \vec{S} \rangle_t \cdot d\vec{a}$$

$$= 0 + 0 + \int_{\phi=0}^{2\pi} \int_{z=0}^1 \left(-\frac{I_0^2}{4\pi^2 R^3 \sigma} \right) R d\phi dz$$

$$= -\frac{I_0^2}{2\pi R^2 \sigma}$$

(1)



Problem 6(a)

$$\vec{p} = a\vec{r}$$

Using CPCs: $\vec{\omega} = \omega \hat{z}$

$$(a) \quad \rho = -\vec{\nabla} \cdot \vec{p} = -\vec{\nabla} \cdot (a\vec{r}) = -2a$$

$$\sigma = \vec{p} \cdot \hat{n} = a\vec{r} \cdot \hat{r}|_{r=R} = aR$$

(1/2)

Total charge per unit length

$$= -2a\pi R^2 + aR 2\pi R = 0$$

Applying Gauss' Law: $\oint \vec{E} \cdot d\vec{S} = Q_{in}/\epsilon_0$

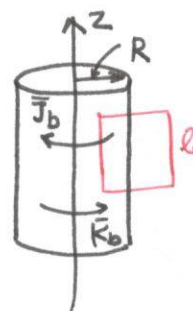
$$\vec{E} = \begin{cases} -\frac{ar}{\epsilon_0} \hat{r} & r < R \\ 0 & r > R \end{cases}$$

(1)

(ii)

$$\vec{J} = \rho \vec{v} = -2a\omega r \hat{\theta}$$

$$\vec{K} = \sigma \vec{v} = aR^2\omega \hat{\theta} \quad (1/2)$$



$\Rightarrow B$ is in the z -direction.

$\Rightarrow$ This is equivalent to the superposition of solenoids $\rightarrow B = 0$ (outside)

Applying Ampere's law

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{in} = \mu_0 \left[\int \vec{J} \cdot d\vec{a} + K\ell \right]$$

$$B\ell = \mu_0 \left[-2a\omega \ell \int_r^R r dr + aR^2\omega \ell \right]$$

$$\Rightarrow B\ell = \mu_0 \left[-\cancel{2}a\omega \ell \frac{(R^2 - r^2)}{\cancel{2}} + aR^2\omega \ell \right]$$

$$\Rightarrow \vec{B} = \begin{cases} \mu_0 a\omega r^2 \hat{z} & \text{inside} \\ 0 & \text{outside} \end{cases}$$

(1)

(iii) ⓐ only electric energy exists if cylinder is not spinning :

$$W_e = \frac{\epsilon_0}{2} \int E^2 dz$$

Energy stored per unit length

$$\begin{aligned} \frac{dW_e}{dz} &= \int_0^{2\pi} d\theta \int_0^R \frac{\epsilon_0}{2} \left(\frac{ar}{\epsilon_0} \right)^2 r dr \\ &= \frac{\pi a^2 R^4}{4 \epsilon_0} \quad \left(\frac{1}{2} \right) \end{aligned}$$

ⓑ After the cylinder's spinning, there exists both the energies - electric as well as magnetic.

$$\begin{aligned} \frac{dW_m}{dz} &= \frac{1}{2\mu_0} \int B^2 dz \\ &= \frac{1}{2\mu_0} \int_0^{2\pi} d\theta \int_0^R (\mu_0 a \omega r^2)^2 r dr \\ &= \frac{\pi \mu_0 a^2 \omega^2 R^6}{6} \quad \left(\frac{1}{2} \right) \end{aligned}$$

∴ Total energy stored per unit length after cylinder starts spinning

$$\frac{dW}{dz} = \frac{dW_e}{dz} + \frac{dW_m}{dz} = \frac{\pi a^2 R^4}{4 \epsilon_0} \left[1 + \frac{2\omega^2 R^2}{3c^2} \right] \quad (1)$$



Problem 6(b)

- ① consider first a uniformly magnetized sphere: using parallels, fields of this sphere are:

$$\vec{H}_{in} = -\frac{1}{3} \vec{M}$$

$$\vec{B}_{in} = \frac{2\mu_0}{3} \vec{M} \quad (1)$$

- ② If this sphere is kept in an external uniform field $\vec{B}_0 = \mu_0 \vec{H}$, then fields inside the sphere are

$$\vec{H}_{in} = \frac{1}{\mu_0} \vec{B}_0 - \frac{1}{3} \vec{M}$$

$$\vec{B}_{in} = \vec{B}_0 + \frac{2\mu_0}{3} \vec{M} \quad (1)$$

- ③ Now suppose that the sphere is made of paramagnetic material (μ). Application of external field will result in finite $\vec{M}$.

Now $\vec{B}_{in} = \mu \vec{H}_{in}$

$$\vec{B}_0 + \frac{2\mu_0}{3} \vec{M} = \mu \left(\frac{1}{\mu_0} \vec{B}_0 - \frac{1}{3} \vec{M} \right)$$

$$\Rightarrow \vec{M} = \frac{3}{\mu_0} \left(\frac{\mu - \mu_0}{\mu + 2\mu_0} \right) \vec{B}_0 \quad (1) \quad \triangle$$

$$7(a) \quad R = \left(\frac{\alpha - \beta}{\alpha + \beta} \right)^2, \quad T = \alpha \beta \left(\frac{2}{\alpha + \beta} \right)^2 \quad (1/2)$$

$$R = T$$

$$\Rightarrow (\alpha - \beta)^2 = 4\alpha\beta$$

$$\Rightarrow \alpha^2 - 6\alpha\beta + \beta^2 = 0$$

$$\therefore \alpha = \frac{6\beta \pm \sqrt{36\beta^2 - 4\beta^2}}{2} = 3\beta \pm 2\sqrt{2}\beta = \alpha \text{ (say)} \quad (1/2)$$

$$\alpha = \frac{\cos \theta_T}{\cos \theta_I} = \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_I \right)^2} \quad (\text{using Snell's law})$$

$$\Rightarrow 1 - \left(\frac{n_1}{n_2} \sin \theta_I \right)^2 = \alpha^2 \cos^2 \theta_I$$

$$\Rightarrow \cos^2 \theta_I = \frac{1 - \left(\frac{n_1}{n_2} \right)^2}{\alpha^2 - \left(\frac{n_1}{n_2} \right)^2} \quad (1)$$

$$\text{For } \mu_1 = \mu_2, \beta = n_2/n_1$$

$$\text{Here } n_1 = 1.0, n_2 = 1.5. \therefore \beta = 1.5 \propto \frac{n_1}{n_2} = \frac{2}{3}$$

$$\text{Let } \alpha = 3\beta - 2\sqrt{2}\beta$$

$$\alpha^2 = (17 - 12\sqrt{2})\beta^2 = 0.0294 \times \frac{9}{4} = 0.067$$

$$\text{Thus, } \alpha^2 - \left(\frac{n_1}{n_2} \right)^2 < 0 \Rightarrow \text{Not acceptable} \quad (1/2)$$

$$\text{So } \alpha = 3\beta + 2\sqrt{2}\beta$$

$$\text{Now } \alpha^2 = (17 + 12\sqrt{2})\frac{9}{4} = 76.43$$

$$\therefore \cos^2 \theta_I = 1 - 4/9 / (76.43 - \frac{4}{9}) = 0.0073$$

$$\Rightarrow \cos^2 \theta_I = 0.0855 \quad \text{or } \theta = 85.09^\circ \quad (1)$$

7(b)

$$R_{\perp} = \left(\frac{\frac{n_1}{\mu_1} \cos \theta_i - \frac{n_2}{\mu_2} \cos \theta_t}{\frac{n_1}{\mu_1} \cos \theta_i + \frac{n_2}{\mu_2} \cos \theta_t} \right)^2 = 0$$

$$= \left(\frac{\frac{n_1}{\mu_1} - \frac{n_2}{\mu_2} \frac{\cos \theta_t}{\cos \theta_i}}{\frac{n_1}{\mu_1} + \frac{n_2}{\mu_2} \frac{\cos \theta_t}{\cos \theta_i}} \right)^2 = \left(\frac{1 - \frac{\mu_1 n_2}{\mu_2 n_1} \frac{\cos \theta_t}{\cos \theta_i}}{1 + \frac{\mu_1 n_2}{\mu_2 n_1} \frac{\cos \theta_t}{\cos \theta_i}} \right)^2$$

$$= \left(\frac{1 - \alpha \beta}{1 + \alpha \beta} \right)^2 = 0 \quad \left(\frac{1}{2} \right)$$

$$\Rightarrow \cos \theta_i = \frac{\mu_1 n_2}{\mu_2 n_1} \cos \theta_t = \frac{2\mu_2 n_1}{\mu_2 n_1} \cos \theta_t$$

$$\Rightarrow \cos \theta_i = 4 \cos \theta_t \quad \text{--- (1)}$$

Using Snell's law

$$n_1 \sin \theta_i = n_2 \sin \theta_t = 2n_1 \sin \theta_t$$

$$\sin \theta_i = 2 \sin \theta_t$$

$$\sqrt{1 - \cos^2 \theta_i} = 2 \sqrt{1 - \cos^2 \theta_t}$$

$$1 - \cos^2 \theta_i = 4 - 4 \cos^2 \theta_t$$

$$\cos^2 \theta_t = \frac{3}{4} + \frac{1}{4} \cos^2 \theta_i$$

$$\cos \theta_t = \sqrt{\frac{3}{4} + \frac{1}{4} \cos^2 \theta_i} \quad \text{--- (2)} \quad (1)$$

Using (1) and (2)

$$\cos^2 \theta_i = 4^2 \left(\frac{3}{4} + \frac{1}{4} \cos^2 \theta_i \right)$$

$$\cos \theta_i = 2$$

$$\theta_i = \cos^{-1}(2)$$



Note: Many students may follow different expressions as given below:

$$R_{11} = \left(\frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t} \right)^2$$

$$T_{11} = \left[\frac{2 n_1 \cos \theta_i}{(n_2 \cos \theta_i + n_1 \cos \theta_t)} \right]^2 \frac{n_2 \cos \theta_t}{n_1 \cos \theta_i} \quad (1/2)$$

Then as $R_{11} = T_{11}$

$$(n_2 \cos \theta_i - n_1 \cos \theta_t)^2 = 4 n_1 n_2 \cos \theta_i \cos \theta_t$$

$$(n_2 \cos \theta_i)^2 + (n_1 \cos \theta_t)^2 - 6(n_2 \cos \theta_i)(n_1 \cos \theta_t) = 0$$

$$x^2 - 6xy + y^2 = 0 \quad (1)$$

$$x = \frac{6y \pm \sqrt{36y^2 - 4y^2}}{2}$$

$$= 3y \pm 2\sqrt{2}y$$

$$\text{case (I)} \quad n_2 \cos \theta_i = (3 + 2\sqrt{2})(n_1 \cos \theta_t) \quad \text{--- (1)}$$

$$\text{case (II)} \quad n_2 \cos \theta_i = (3 - 2\sqrt{2})(n_1 \cos \theta_t) \quad \text{--- (2)}$$

$$\text{But} \quad n_1 \sin \theta_i = n_2 \sin \theta_t \quad (1)$$

$$\sin \theta_t = \frac{n_1}{n_2} \sin \theta_i$$

$$\Rightarrow \cos \theta_t = \sqrt{\frac{4}{9} \cos^2 \theta_i + \frac{5}{9}} \quad \text{--- (3)}$$

(1) and (2) will not give a possible θ_i (1/2)

(2) and (3) will give $\theta_i \approx 85.09^\circ$ (1)

